

Autoformer for Long-Term Time Series Forecasting on ETTm2: Architecture and Proposed Modifications



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Time Series & Benchmarks

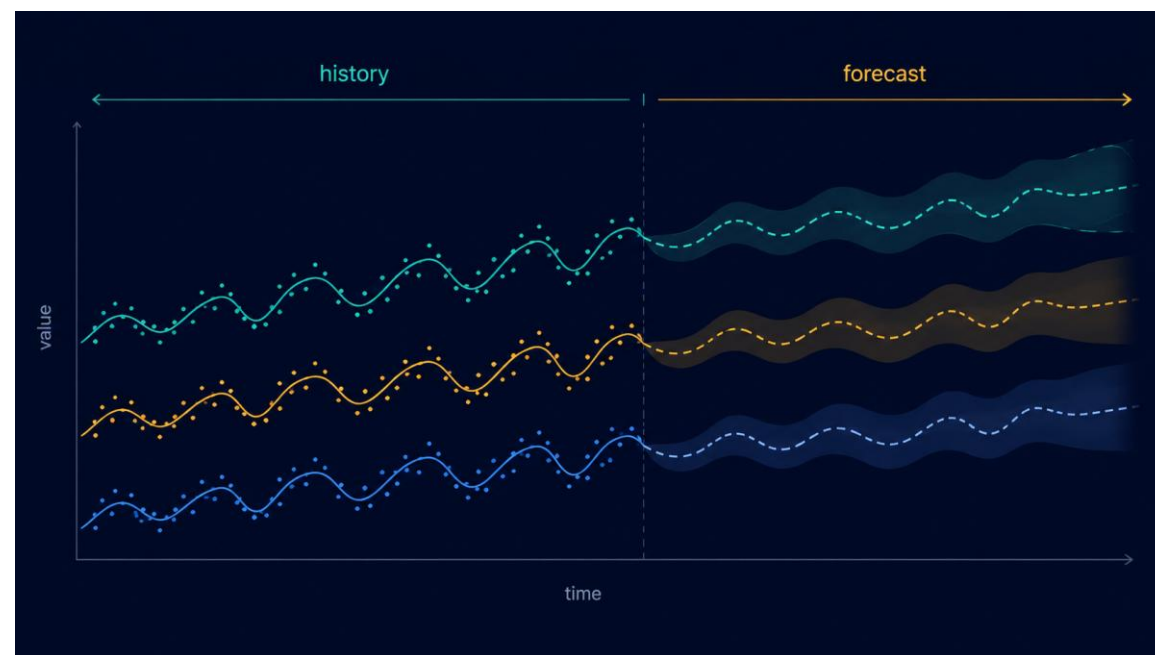
Introduction to Time Series

Predicting the Future

Time series forecasting involves analyzing sequences of data points indexed in time order to predict future values.

Applications range from energy planning to disease propagation.

- The Challenge: Long-term forecasting is difficult due to intricate, entangled temporal patterns. Forecasting longer horizons usually amplify uncertainty and error accumulation.
- The Bottleneck: Traditional Transformers rely on point-wise self-attention, creating a quadratic complexity barrier $O(L^2)$ for long sequences.
- The Goal: Discovering reliable long-range dependencies efficiently.



ETTm2 Benchmark Dataset



Domain & Context

ETT stands for Electricity Transformer Temperature. It tracks crucial indicators of electric power deployment and transformer condition, highly relevant for real-world energy infrastructure planning.



Resolution & Span

The ETTm2 dataset is recorded at a 15-minute level over a span of two years. This dense, high-frequency data provides a rigorous testbed for evaluating long-term forecasting capabilities.



Multivariate Features

It contains multiple target variables, specifically oil temperature (a key safety metric) alongside 6 distinct power load features, demanding models that can handle complex cross-channel interactions.

ETTm2 Benchmark Dataset

	date TIMESTAMP: 15-MINUTE GRANULARITY		HUFL HIGH USEFUL LOAD HIGH VOLTAGE (Primary Winding): ACTIVE POWER (Fuel-Useful Load) Units: e.g., kW
	HUFL HIGH USEFUL LOAD HIGH VOLTAGE (Primary Winding): ACTIVE POWER (Fuel-Useful Load) Units: e.g., kW		HULL HIGH USELESS LOAD HIGH VOLTAGE (Primary Winding): REACTIVE POWER (Load-Useless Load) Units: e.g., kVAR
	MUFL MEDIUM USEFUL LOAD HIGH VOLTAGE (Primary Winding): ACTIVE POWER (Fuel-Useful Load) (Secondary Winding 1)		LULL LOW USELESS LOAD LOW VOLTAGE (Primary Winding): LOW VOLTAGE (Secondary Winding 2) Units: e.g., kVAR
	LUFL LOW USEFUL LOAD LOW VOLTAGE (Primary Winding): (Secondary Winding 2) (Secondary Winding 2)		OIL TEMPERATURE TARGET VARIABLE: OIL TEMPERATURE Units: e.g., °C

Autoformer Architecture

Auto-Correlation

Replace self-attention

- Instead of asking which individual time points are related, discover periodic/time-delay relationships between sub-series.
- Use those repeated patterns to aggregate information.
- We are going to correlating time series with itself

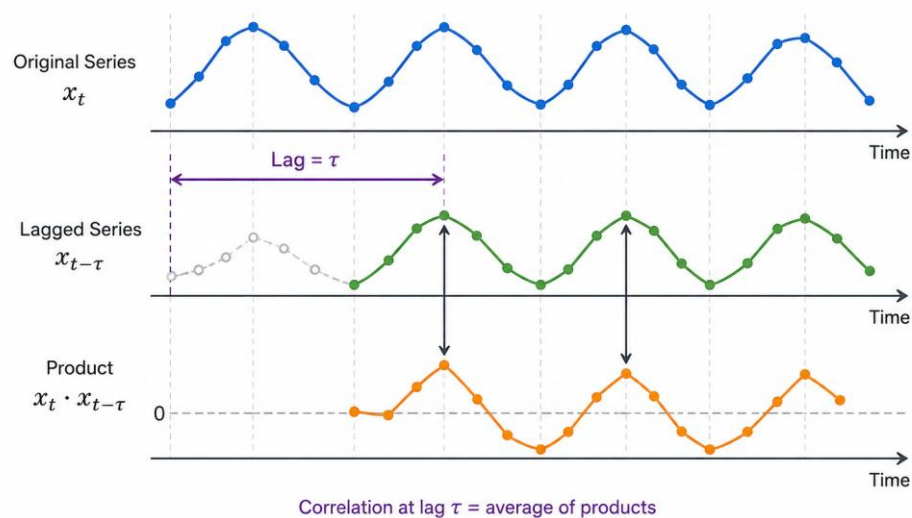
What does auto-correlation mathematically measure?

At its simplest way we are asking:

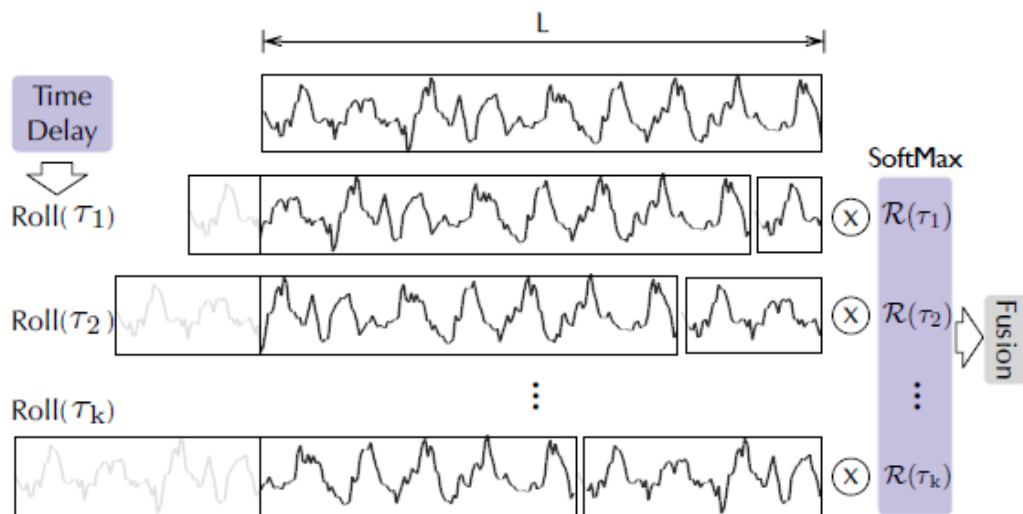
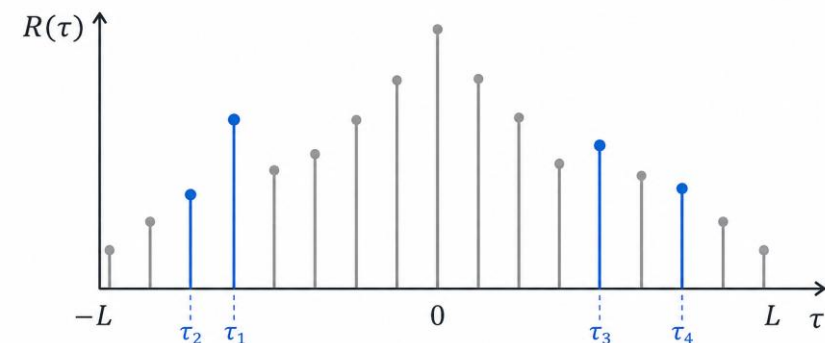
When the series is shifted by τ , how similar is it to the original series?

$$R(\tau) = \sum_t X_t X_{t-\tau}$$

Lag in Auto-Correlation



$$R(\tau) = \sum_t X_t X_{t-\tau}$$



$$\tau_1, \dots, \tau_k = \arg \text{Topk}_{\tau \in \{1, \dots, L\}} (R_{Q,K}(\tau)), \quad k = \lfloor c \cdot \log L \rfloor$$

$$\hat{R}_{Q,K}(\tau_1), \dots, \hat{R}_{Q,K}(\tau_k) = \text{SoftMax}(R_{Q,K}(\tau_1), \dots, R_{Q,K}(\tau_k))$$

$$\text{Auto-Correlation}(Q, K, V) = \sum_{i=1}^k \text{Roll}(V, \tau_i) \cdot \hat{R}_{Q,K}(\tau_i)$$

FFT (Fast Fourier Transform) Utilization

Instead of calculating:

$$R(0), R(1), R(2), \dots$$

one at a time in the time domain, we can move to the frequency domain.

And there is a mathematical theorem (Wiener–Khinchin theorem) that says:

Autocorrelation in the time domain can be obtained from multiplication in the frequency domain.

FFT (Fast Fourier Transform) Utilization

For a sequence of length L:

$$x_0, x_1, \dots, x_{L-1},$$

The DFT produces:

$$X_0, X_1, \dots, X_{L-1}.$$

The formula is:

$$X_k = \sum_{n=0}^{L-1} x_n e^{-i2\pi kn/L}$$

$$R_{xy}[\tau] = \sum_t x[t] y[t - \tau] = (x * y_{\text{rev}})[\tau]$$

$$y_{\text{rev}}[t] = y[-t]$$

Apply the convolution theorem:

$$\mathcal{F}\{x * y_{\text{rev}}\} = \mathcal{F}\{x\} \mathcal{F}\{y_{\text{rev}}\}$$

For a real-valued sequence y[t]:

$$\mathcal{F}\{y[-t]\} = Y[-k] = Y^*[k]$$

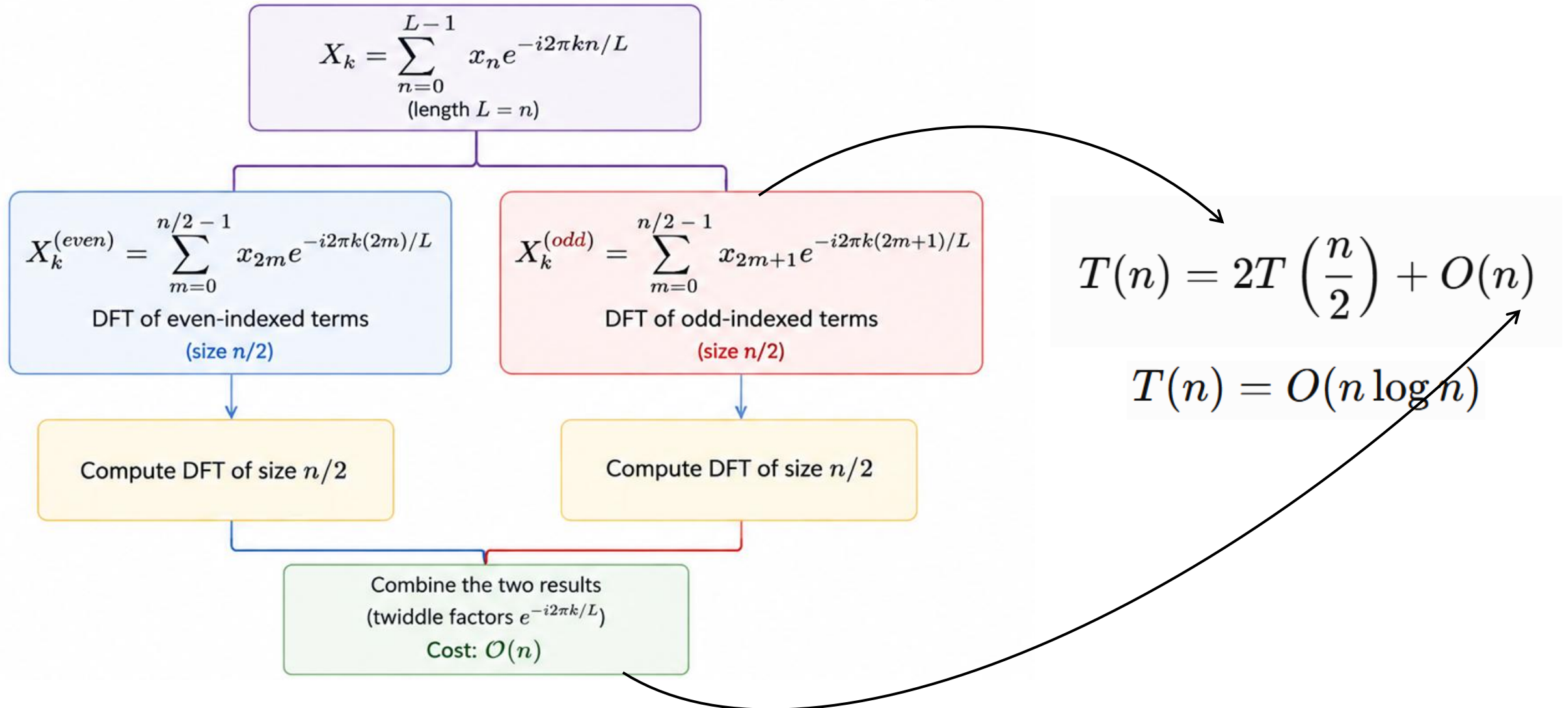
Therefore:

$$\mathcal{F}\{R_{xy}[\tau]\} = X[k]Y^*[k]$$

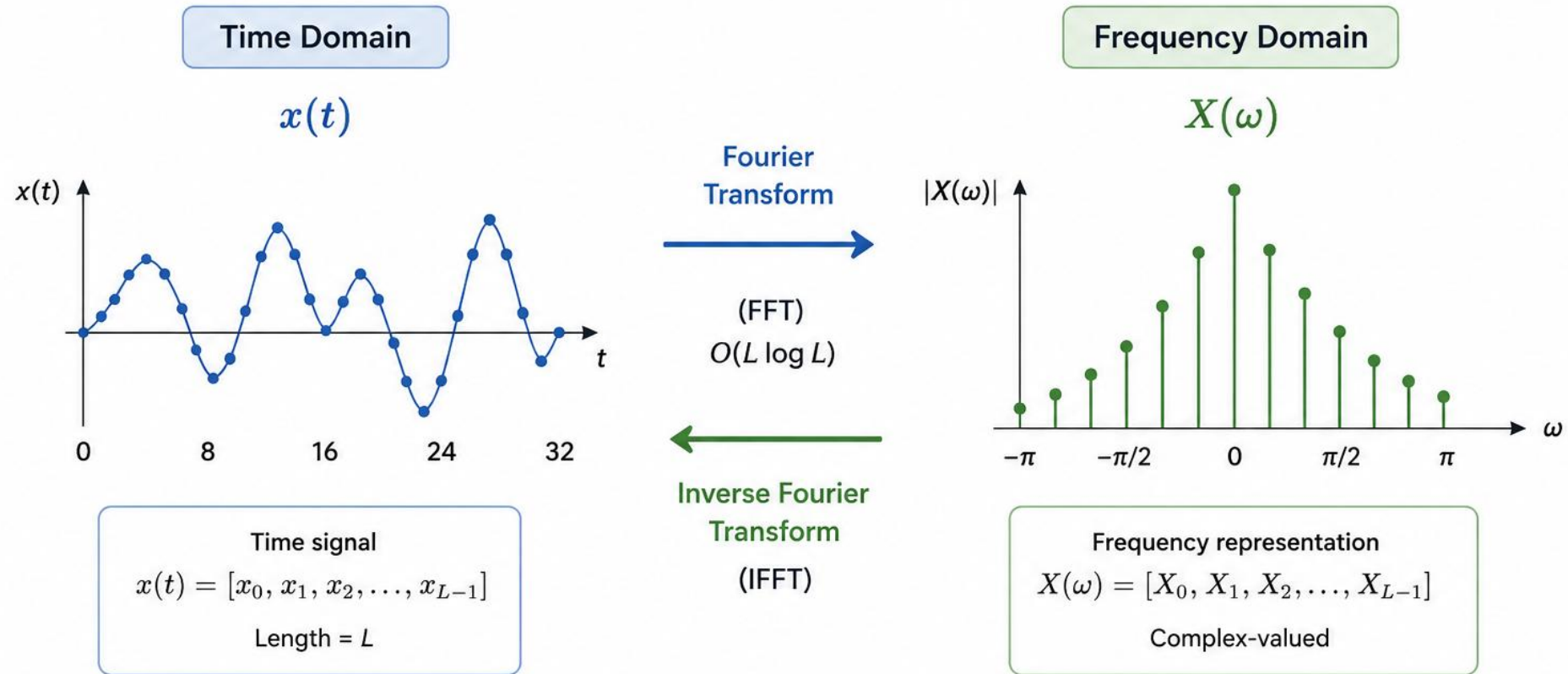
and going back to the time domain:

$$R_{xy}[\tau] = \text{IFFT}(X[k]Y^*[k])$$

FFT (Fast Fourier Transform) Utilization



FFT (Fast Fourier Transform) Utilization

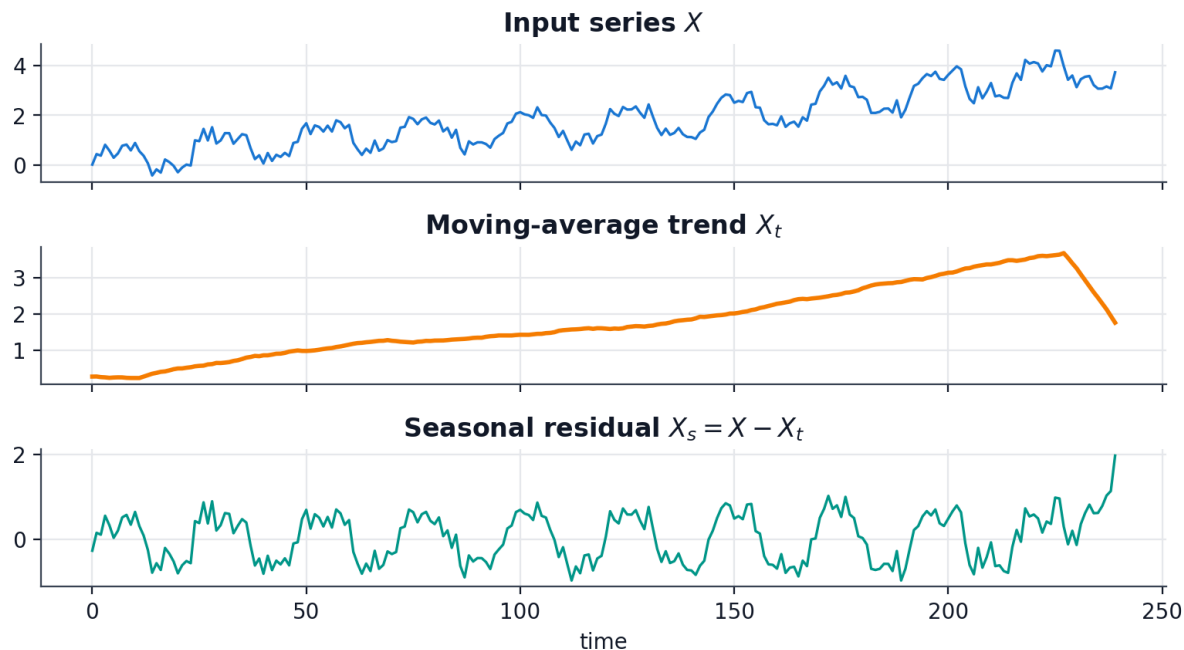


Idea

Time signal $x(t)$ $\xrightarrow{\text{FFT}}$ Frequency spectrum $X(\omega)$
(how it changes over time) Convert (how much of each frequency is present)

Decomposition

Original Autoformer decomposition



one fixed moving-average kernel is used as the decomposition scale.

A time series usually contains different types of behavior mixed together. Autoformer separates them into two main components:

$$X = \text{Trend} + \text{Seasonal}$$

- Trend: the slowly changing, long-term direction of the series.
- Seasonal: the faster fluctuations and repeating patterns around the trend.

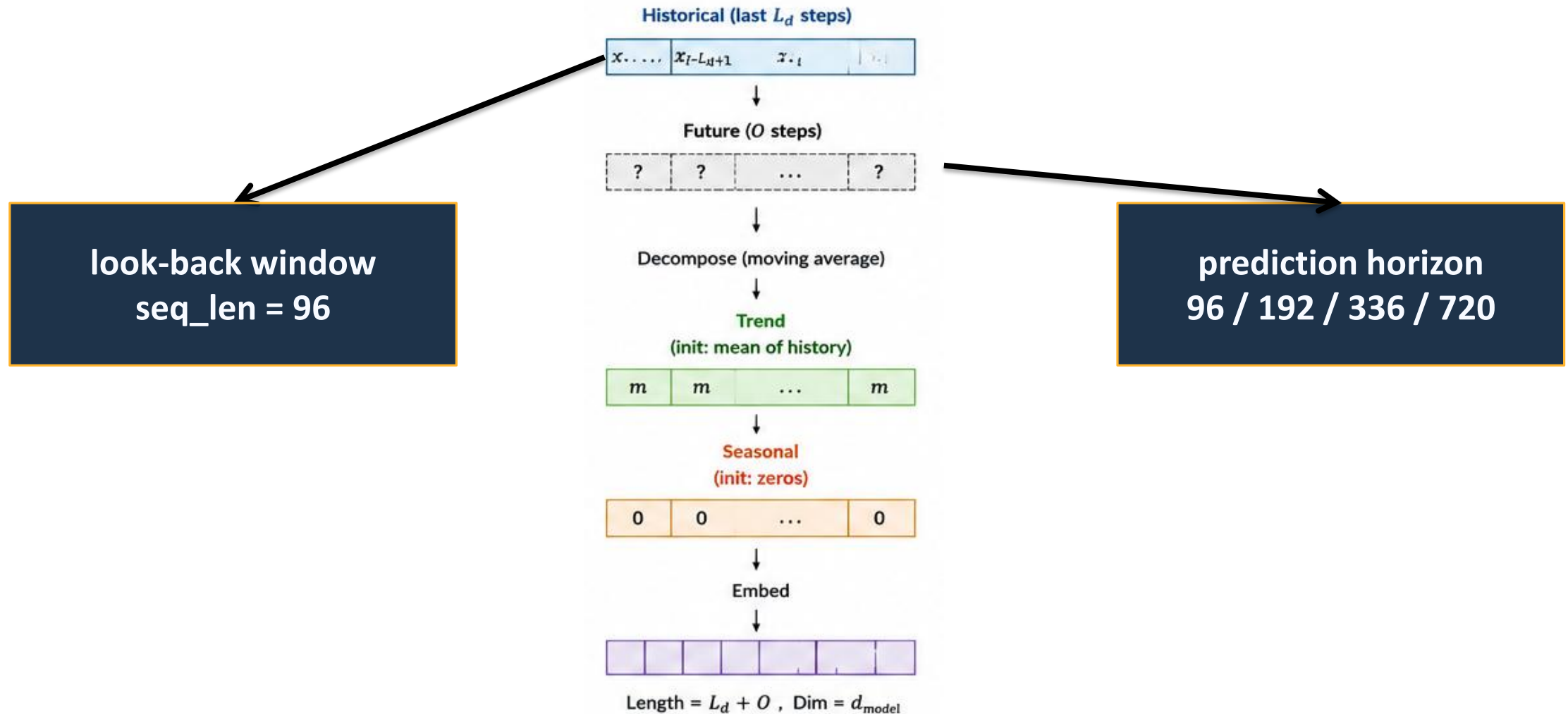
$$X \longrightarrow \begin{cases} X_{\text{trend}} \\ X_{\text{seasonal}} \end{cases}$$

$$X_{\text{trend}} = \text{MovingAvg}(X)$$

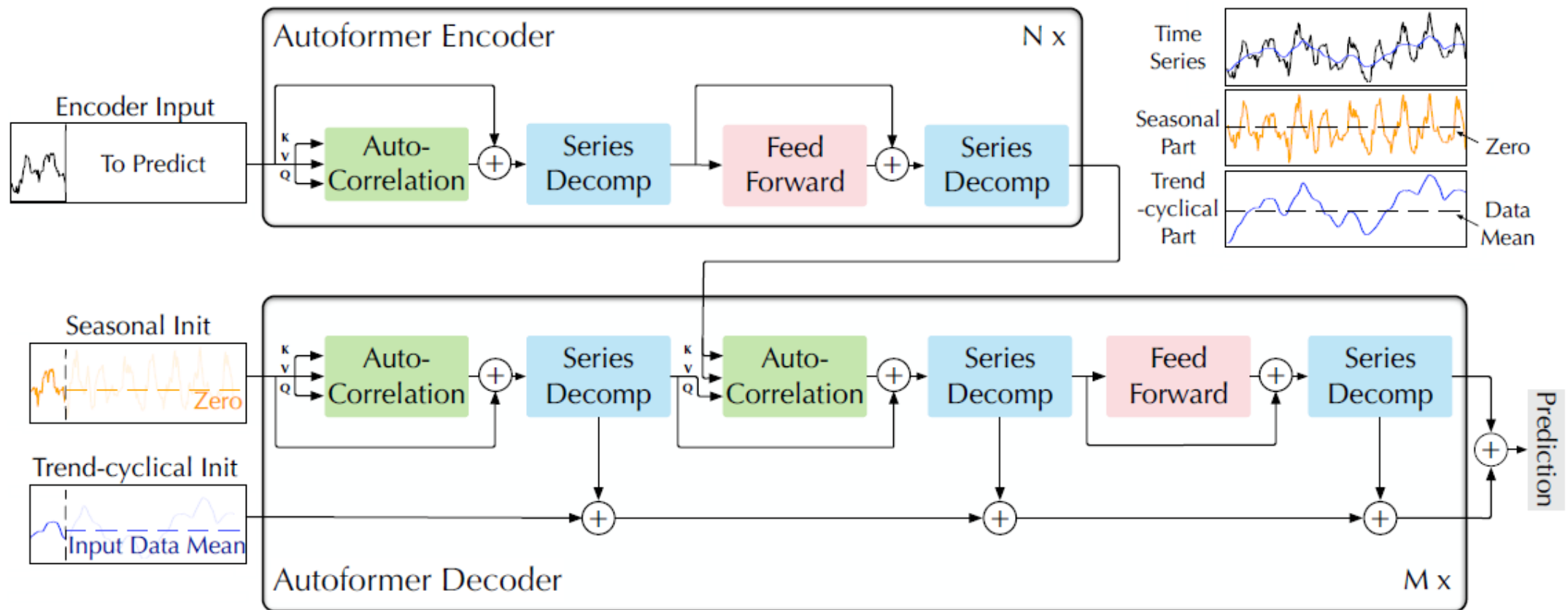
$$X_{\text{seasonal}} = X - X_{\text{trend}}$$

Decoder Input Format

1. Decoder Input Construction



Autoformer Encoder-Decoder



Architectural Modifications(1)

Adaptive Multi-Scale Decomposition

Original Approach (Fixed Scale)

The baseline Autoformer uses a single, fixed moving average window to extract the trend. This forces one smoothing scale to serve all layers and varying input representations.

$$T = MA(X)$$

Multi-Scale Adaptive Fusion

The proposed modification utilizes multiple temporal scales (e.g., short, medium, long) and learns fusion weights to dynamically blend the optimal trend configuration.

$$T = \alpha_1 MA_{k_1}(X) + \alpha_2 MA_{k_2}(X) + \alpha_3 MA_{k_3}(X)$$

$$\alpha_1 + \alpha_2 + \alpha_3 = 1.$$

Adaptive Multi-Scale Decomposition

Why this modification is valuable?

More flexible trend extraction

a short window captures local trend changes; a long window captures slow operating regimes

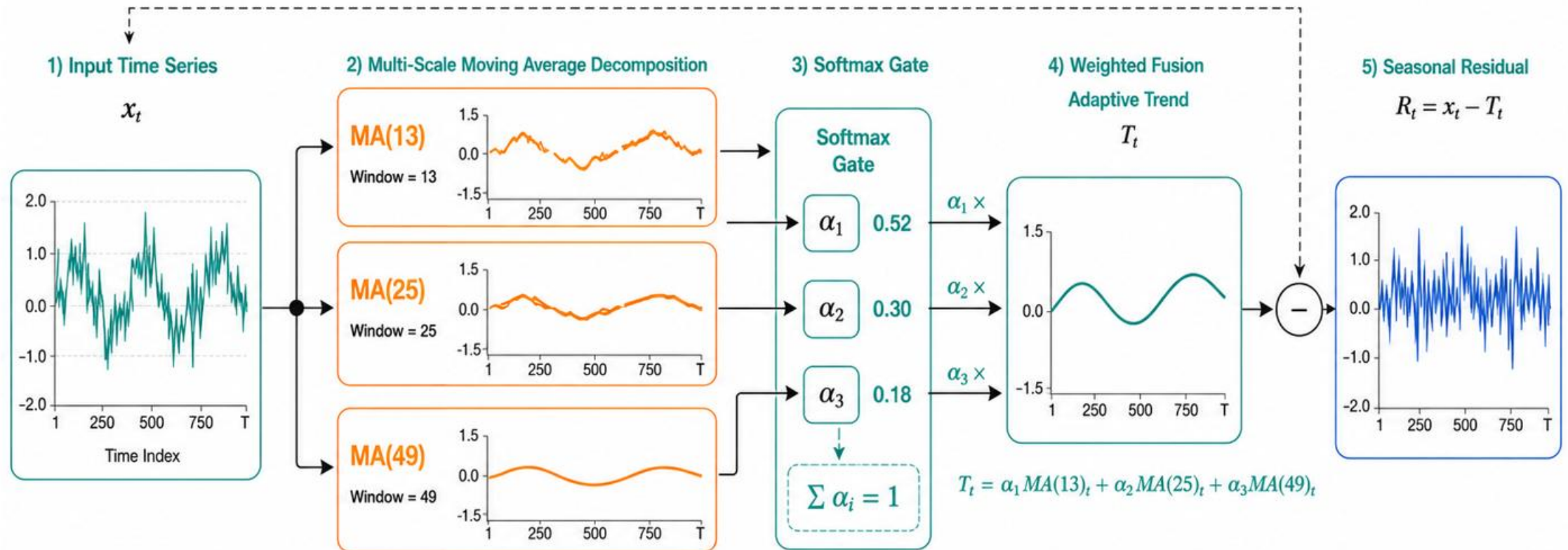
Channel-specific behavior

ETM2 variables may have different smoothness and periodicity; one window may not fit all

Preserves Autoformer logic

decomposition remains progressive; Auto-Correlation and decoder trend accumulation stay intact

Adaptive Multi-Scale Decomposition



Architectural Modifications(2-A)

Sample-Adaptive Lag Selection

Limitation of Original Autoformer

- In Autoformer, the Auto-Correlation mechanism first computes the correlation strength of different time delays for every input sample.
- However, in the training speed-up version, Autoformer averages these autocorrelation values across the samples in the batch before selecting the Top-k delays.

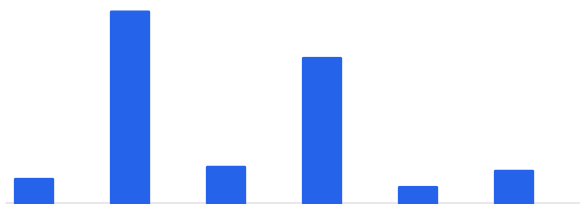
$$R_{\text{batch}}(\tau) = \frac{1}{B} \sum_{b=1}^B R_b(\tau)$$

- Different time-series windows may contain different dominant periodicities, but batch-level lag selection can hide these sample-specific patterns.

Sample-Adaptive Lag Selection

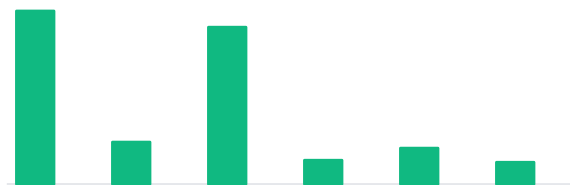
averaging can hide sample-specific periods

Sample A wants lags {2,4}



A loses lag 4

Sample B wants lags {1,3}



B loses lag 1

Batch average chooses {2,3}



Both use same lags

Sample-Adaptive Lag Selection

The main idea of Experiment 2A is simple:

Do not average the autocorrelation across the batch before selecting Top-k delays.

Instead of:

$$[B, L] \rightarrow \text{Mean over Batch} \rightarrow [L] \rightarrow \text{Top}K$$

we use:

$$[B, L] \rightarrow \text{Top}K \rightarrow [B, K]$$

Now every sample selects its own most important delays

Architectural Modifications(2B)

Adaptive Period Routing

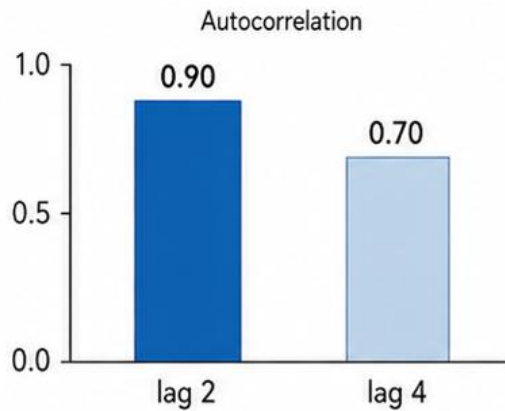
- Correlation Strength Is Not Always Forecasting Importance.
- After Autoformer selects the Top-k delays, it determines their importance mainly from their Auto-Correlation values.

$$\alpha_i = \text{Softmax} (R(\tau_i))$$

Limitation: A period can be strongly correlated with the past, but this does not necessarily mean that it is the most useful period for predicting the future.

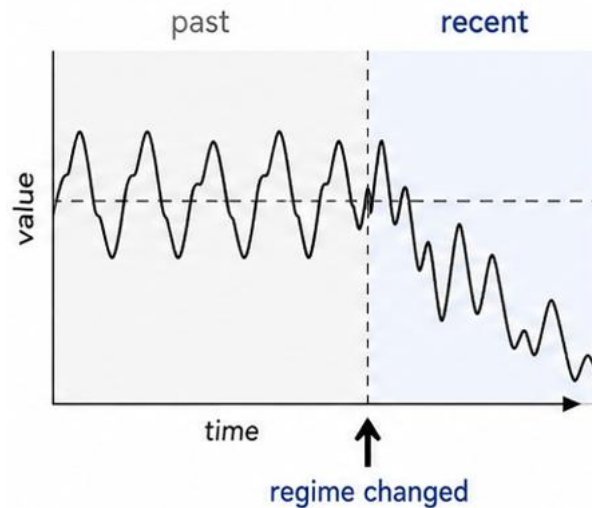
Adaptive Period Routing

For Sample A:



Autocorrelation says:
lag 2 > lag 4

By autocorrelation alone, lag 2
would receive more weight.



The most recent behavior is different
from the historical pattern.



Because recent behavior
changed, lag 4 provides better
forecasting information, which
is in paradox of having less
Auto-correlation score than
lag 2

Adaptive Period Routing

- This experiment introduces a small learnable router (MLP) after the candidate periods have been identified.
- Instead of using only $R(\tau_i)$, to determine the final importance of a period, we calculate:

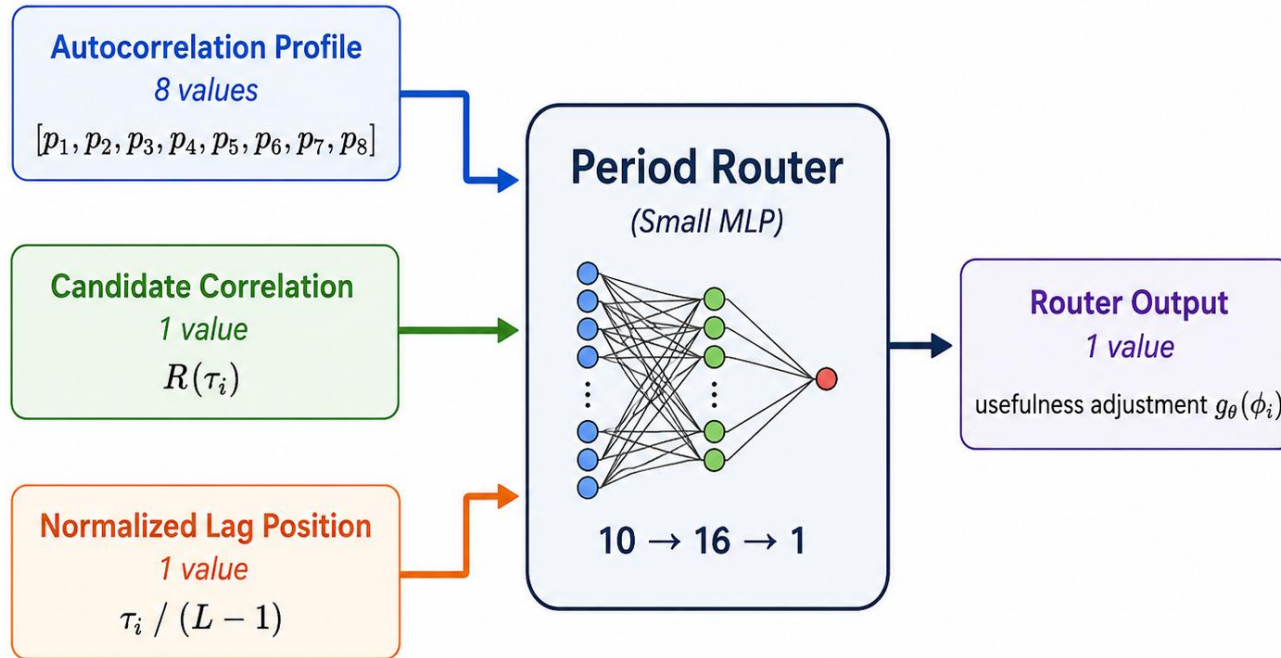
$$s_i = R(\tau_i) + g_{\theta}(\phi_i)$$

where:

- $R(i)$ original Auto-Correlation strength
 - g_{θ} : small neural-network router
 - ϕ_i : information describing the current sample and candidate period.
-
- The final weights become:

$$\alpha_i = \text{Softmax}(s_i)$$

Adaptive Period Routing



MLP Input = 8 + 1 + 1 = 10 values

The same router evaluates each candidate lag separately.

For each candidate lag, the MLP receives **10 input values**:

- **8 values** summarize the overall autocorrelation pattern of the current sample.
- **1 value** represents the autocorrelation strength of the candidate lag.
- **1 value** represents the normalized position of that lag in the sequence.

Experiments

These two ideas:

- Sample-Adaptive Lag Selection (2A)
- Adaptive Period Routing (2B)

are going to be used in two different situation. The first experiment just used 2A but the second one uses both approach simultaneously

Results

Results of Experiments

Horizon	Baseline MSE/MAE	Adaptive multi-scale decomposition MSE/MAE	2A MSE/MAE	2A+2B MSE/MAE
96	0.247/0.324	0.239/0.316	0.221/0.307	0.232/0.313
192	0.282/0.341	0.283/0.345	0.280/0.340	0.272/0.330
336	0.331/0.367	0.404/0.406	0.331/.366	0.323/0.360
720	0.435/0.429	0.476/0.450	0.425/0.419	0.423/0.418

Thank You
